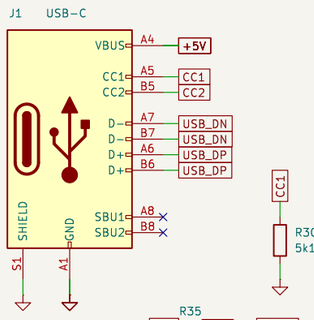
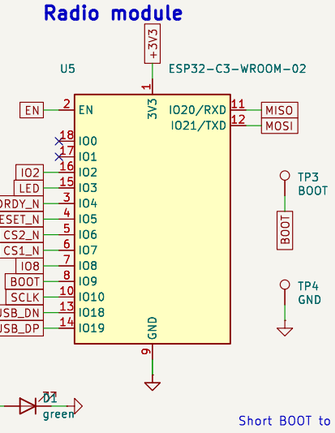
NOT A MEDICAL DEVICE. No isolation: run from a battery / power bank only while worn.

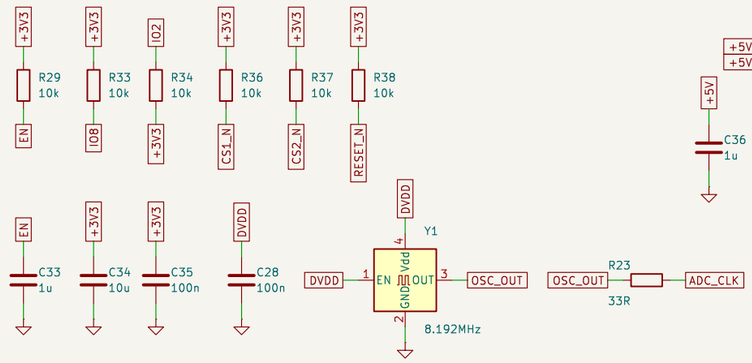
USB-C: power and flashing



Radio module



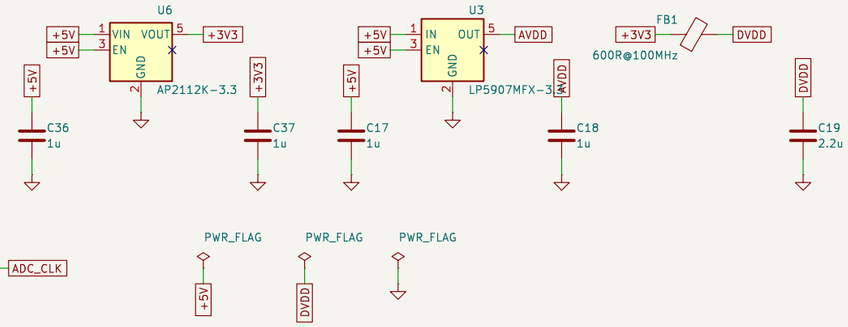
Enable, boot straps, pull-ups



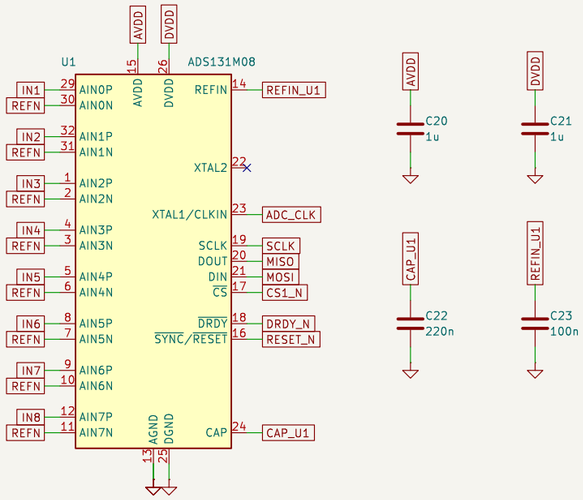
Short BOOT to GND while plugging in USB to force download mode.

Clock: 8.192 MHz to both converters

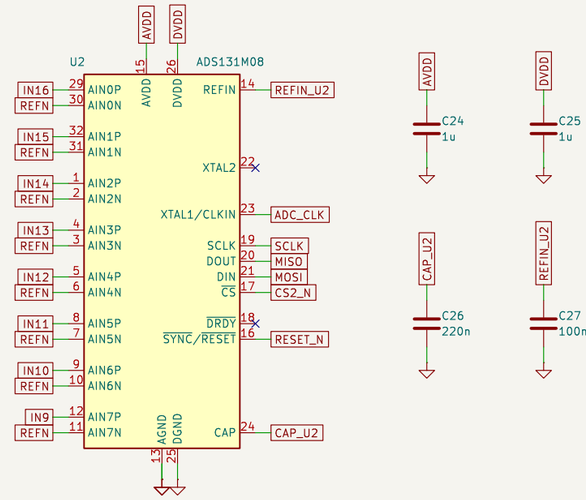
Power: 5 V in, 3.3 V for the module, quiet 3.3 V for the analog side



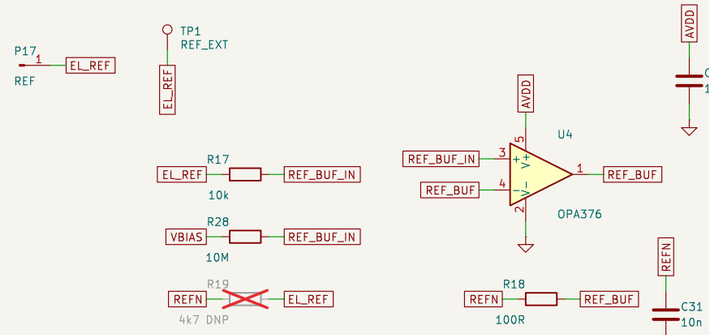
ADC A: channels 1-8



ADC B: channels 9-16

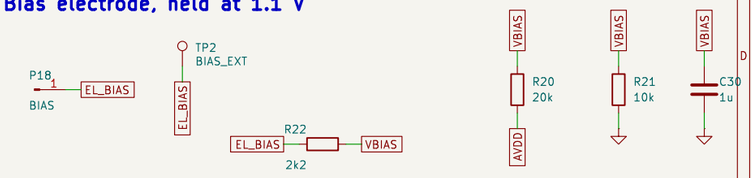


Reference electrode and unity-gain buffer



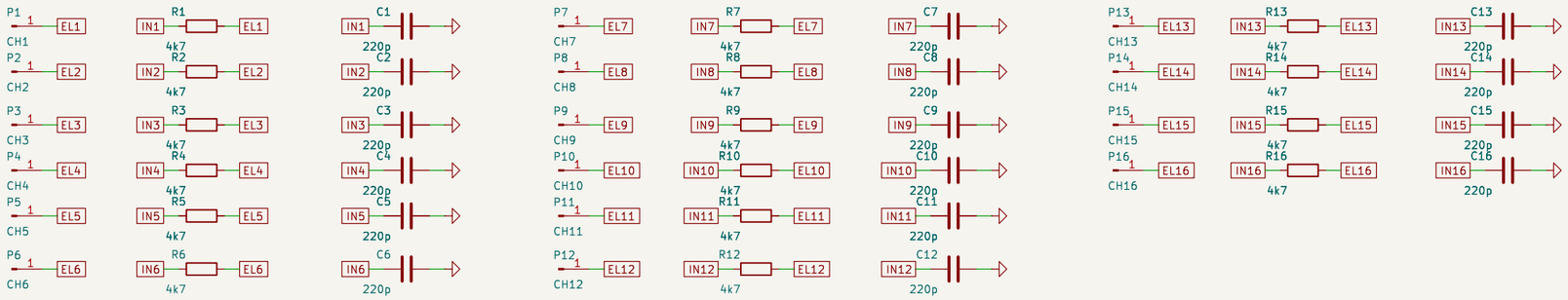
R18 fitted, R19 not fitted: buffered reference (default).
R19 fitted, R18 removed, C31 changed to 220 pF: electrode drives the reference node directly.

Bias electrode, held at 1.1 V



Remove R20 to hold the bias electrode at ground (then use the direct reference: the buffer needs the 1.1 V bias to work).

Electrodes and input filters: pogo pin -> 4k7 -> ADC input, 220 pF to ground



Untested revision. Not a medical device.
Generated from hardware/scripts/design.py. Edit that file, not this one.
Open hardware, CERN-OHL-P-2.0

Sheet: /
File: squid16.kicad_sch

Title: Squid-16: 16-channel dry-electrode EEG front end

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 9.0.9.1		Id: 1/1